DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2015/2016

THIRD YEAR SEMESTER TWO EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE / BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY / BACHELOR OF BUSINESS INFORMATION TECHNOLOGY

ICS 2311: COMPUTER GRAPHICS

DATE: 23RD December 2015 TIME: 4.00pm-6.00pm

INSTRUCTIONS: ANSWER QUESTION 1 AND ANY OTHER TWO.

QUESTION ONE (30 MARKS)

a) Explain the following terms
   i) Persistence [2 marks]
   ii) Scan code [2 marks]
   iii) Addressability [2 marks]
   iv) Random scan display [2 marks]
   v) Interdot distance [2 marks]

b) List any three merits of electrostatic printers [3 marks]

c) Differentiate emissive from non emissive displays [4 marks]

d) Differentiate scaling from shearing as applied in computer graphics [4 marks]

e) Explain the following OpenGL functions [3 marks]
   i) glVertex2i(12,30)
   ii) glClearColor (1.0,1.0,0.7,0.1)
   iii) glDisplayMode(object)

f) Differentiate concave from convex polygons [2 marks]

g) State and explain any two representation schemes used in 3D objects [4 marks]

QUESTION TWO (20 MARKS)

a) Using appropriate diagrams, differentiate interlacing scan from progressive scan [4 marks]

b) Using examples, differentiate lossy from non-lossy images [4 marks]

c) Give a common pixel data format that uses up the following numbers of bits per pixel: 8, 16, 32, 36. For instance, given 24 bits per pixel you might answer “8 bits for each of red, green, and blue.” [4 marks]

d) Explain the need of homogeneous coordinate transformation [4 marks]

e) Explain an advantage of a display system that uses nonlinear intensity quantization (that is, one that requires gamma correction) over a display that uses linear quantization, given that the purpose of the display is to show images to a human viewer.

[4 marks]

QUESTION THREE (20 MARKS)

a) Distinguish between window port & view port [4 marks]
b) Homogeneously translate point A (3,5) by factors (1,2) [3marks]
b) Describe covering (exterior clipping) [3 marks]
c) Using an example, describe Bresenham line drawing algorithm [10 marks]

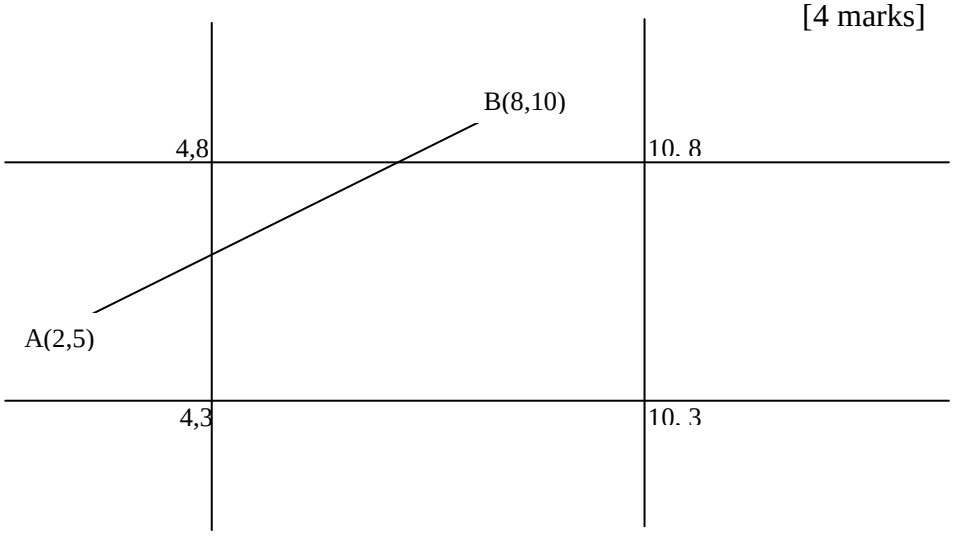
QUESTION FOUR (20 MARKS)

a) Identify and discuss any four graphics output primitives [6 marks]
b) Identify and discuss any four OpenGL functions [6 marks]
c) How can Dedan Kimathi University of Technology apply computer graphics for learning purposes? [4 marks]
d) Suppose you are using a reconstruction filter for translating images around in the plane without changing their size, and it performs well for that application. For each of the following other applications, explain if it is necessary to make the filter larger, make it smaller, or keep it the same size relative to the input pixel grid.

i) Reducing the image (down-sampling) [2 marks]
ii) Enlarging the image (up-sampling) [2 marks]

QUESTION FIVE (20 MARKS)

a) Describe Affine transformation [4 marks]
b) Rotate point A(8,7) through angle of 45^0 about an arbitrary point B(2,3) [4 marks]
c) Explain the various Text clipping techniques [6 marks]
d) Compute the points of intersections of the following clipping region with line AB [4 marks]



e) What are the important properties of Bezier Curve [2 marks]